

Why the Analytic Core Resists Formalization: A Rigorous Characterization of Three Irreducible Lemmas in the Tao–Collatz Development

Aristotle

(companion note to `RequestProject/TaoCollatz.lean`)

July 7, 2026

Abstract

The machine-checked development accompanying this note reduces Tao’s Theorem 1.3 (“almost all Collatz orbits attain almost bounded values”) to exactly three explicitly-typed, non-vacuous sorry milestones: `step4_valuationLowerBound`, `step6_typicalDescent`, and `step7_oddControl`. Everything else — the elementary dynamics, the odd-part/Syracuse simulation, the exact affine backbone, the exact 2-adic valuation distribution, and the whole reduction chain — is proved without axioms. This note gives a *rigorous, formal* account of *why* these three remaining lemmas are hard to formalize. We do not merely assert that they are “deep”; we isolate, for each, a precise structural obstruction that survives even after all the surrounding arithmetic has been discharged. The obstructions are of three genuinely distinct kinds: (i) a *deterministic-to-probabilistic transfer* that has no counterpart in the current Mathlib library and cannot be an equality of finite objects (step 4); (ii) a *provable failure of pointwise algebraic reducibility*: the additive correction term in the affine backbone is not controlled by the multiplicative contraction hypothesis, so the reduction is not a rewriting step but requires the distribution (step 6); and (iii) a *self-referential stability / renewal* requirement whose density hypothesis must be inherited by the orbit’s own descendants (step 7). We make each claim precise as a stated proposition or obstruction, quote the exact Lean goals, and enumerate the missing library infrastructure.

1 Setup: the objects, and what is already proved

We recall only what is needed to state the obstructions rigorously. All identifiers below are the actual declarations in `RequestProject/TaoCollatz.lean` (namespace `TaoCollatzLean`); the same content is mirrored in the Idris2 tree under `TaoCollatz/`.

1.1 The maps

The Collatz map is `collatz n = if n % 2 = 0 then n/2 else 3*n+1`. The *odd part* `odd(n)` (Lean `oddFactor`) is the largest odd divisor of n , and the *Syracuse map* on odd numbers is

$$\text{Syr}(x) = \text{odd}(3x + 1) \quad (\text{Lean } \text{syrr}).$$

The development proves outright (no hypotheses) the two facts that make the Syracuse map the right object of study:

Lemma 1.1 (dynamics, proved). *collatz_drop*: iterating `collatz` from n reaches `odd(n)`; and *colBelow_of_syrBelow*: if the Syracuse orbit of `odd(n)` eventually drops $\leq b$, so does the Collatz orbit of n .

Thus the whole problem is transported, with a fully verified simulation, to the Syracuse map. The remaining difficulty is entirely about `Syr`.

1.2 The valuation and the exact affine backbone

Write $\mathbf{a}(x)$ (Lean `syrVal x`) for the number of halvings in one Syracuse step, i.e. the 2-adic valuation of $3x + 1$, and let

$$\mathbf{S}_m(x) = \mathbf{a}(x) + \mathbf{a}(\text{Syr}x) + \cdots + \mathbf{a}(\text{Syr}^{m-1}x) \quad (\text{Lean } \text{syrValSum } m \ x).$$

The single most important *proved* algebraic identity is the affine backbone:

Lemma 1.2 (`affineBackbone`, proved). *For all x, m there is $c_m = c_m(x) \geq 0$ with*

$$2^{\mathbf{S}_m(x)} \text{Syr}^m(x) = 3^m x + c_m(x).$$

Moreover the proof exhibits the recursion $c_0 = 0$ and $c_{m+1}(x) = 3^m + 2^{\mathbf{a}(x)} c_m(\text{Syr}x)$.

This identity is exact and elementary; it is the load-bearing bridge between the multiplicative data ($\mathbf{a}$) and the actual orbit values. The reader should keep the explicit form of c_m in mind: *it depends on the entire valuation profile* $(\mathbf{a}(x), \dots, \mathbf{a}(\text{Syr}^{m-1}x))$, not just on the sum $\mathbf{S}_m$. This is the crux of §4.

1.3 The exact valuation distribution (proved)

The development also proves the exact combinatorics of the valuation, i.e. the finitary heart of Tao’s Proposition 1.9:

Proposition 1.3 (`syrVal_tail_count_period`, `syrVal_level_count_period`, proved). *For $1 \leq j \leq K$, exactly 2^{K-j} residues $x \in \{0, \dots, 2^K - 1\}$ have $\mathbf{a}(x) \geq j$; and for $1 \leq j, j+1 \leq K$, exactly 2^{K-j-1} residues have $\mathbf{a}(x) = j$ exactly. Equivalently, under the uniform measure on residues mod 2^K ,*

$$\mathbb{P}[\mathbf{a} \geq j] = 2^{-j}, \quad \mathbb{P}[\mathbf{a} = j] = 2^{-(j+1)}.$$

This is a statement about *one* step and *finitely many* residues, and it is decidable in principle; that is exactly why it *is* proved. Note $\mathbb{E}[\mathbf{a}] = \sum_{j \geq 1} \mathbb{P}[\mathbf{a} \geq j] = \sum_{j \geq 1} 2^{-j}$?; more precisely, from the geometric law $\mathbb{P}[\mathbf{a} = j] = 2^{-(j+1)} \cdot 2 = 2^{-j}$ on odd residues one gets the mean drift $\mathbb{E}[\mathbf{a}] = 2$, comfortably above the critical rate $8/5 = \log_2 3$ needed to force contraction (since $3^5 < 2^8$, proved as `strictGrowth`). The gap between this per-step, finitary statement and the three milestones below is the entire subject of this note.

1.4 The three milestones

The reduction chain is

$$\underbrace{\text{step 4}}_{\text{drift}} \xrightarrow{\text{step 5 (proved)}} \underbrace{\text{contraction}} \xrightarrow{\text{step 6}} \underbrace{\text{typical descent}} \xrightarrow{\text{step 7}} \underbrace{\text{first passage}}_{\text{SyracuseDensityControl}} \Rightarrow \text{Thm 1.3}.$$

Step 5 (`step5_contractionFromValuation`) is fully proved: it is the purely arithmetic passage from the drift $8m \leq 5\mathbf{S}_m$ to $3^m f(n) \leq 2^{\mathbf{S}_m}$, via `contractionArith`. The three unproved sorry milestones are, verbatim:

```
def ValuationLowerBoundDensity : Prop :=
  f, TendsToInfinity f → good, DensityZero (fun n => !good n)
  n, good n = true →
    m, f n m 8 * m 5 * syrValSum m (oddFactor n)
```

```

theorem step4_valuationLowerBound : ValuationLowerBoundDensity := sorry

def TypicalDescentDensity : Prop :=
  good, DensityZero (fun n => !good n)
  n, good n = true → m, 0 < m  syr^[m] (oddFactor n) < oddFactor n
theorem step6_typicalDescent :
  ContractionDominatesDensity → TypicalDescentDensity := sorry

theorem step7_oddControl :
  TypicalDescentDensity → SyracuseDensityControl := sorry

```

Here `DensityZero` `bad` is the genuine natural-density notion $\frac{1}{N} \#\{n < N : \text{bad } n\} \rightarrow 0$, and `TendsToInfinity` `f` is $f \rightarrow \infty$. We stress that none of these is weakened to `True`, and each is non-vacuous (the good set is required to be genuinely large, and step 6’s descent is required to be strict and at a positive time).

2 A common root: deterministic arithmetic behaving probabilistically

Before treating the milestones individually, we isolate the obstruction they share, because it explains why *none* of them is a finite computation.

The valuations $\mathbf{a}(\text{Syr}^i x)$ are *deterministic* functions of the integer x . Yet every heuristic and every known partial result treats the sequence $(\mathbf{a}(\text{Syr}^i x))_{i \geq 0}$ as if it were a sequence of independent Geometric(1/2) random variables. Proposition 1.3 is the rigorous meaning of “ $\mathbf{a}$ is Geometric”: it holds *on average over residues*, not for a fixed x . The transfer principle one needs is:

as x ranges over a residue class mod 2^K with K large, the joint law of $(\mathbf{a}(x), \dots, \mathbf{a}(\text{Syr}^{m-1} x))$ converges (with a usable rate) to the product geometric law, uniformly enough to conclude a density-one statement about the integers n themselves.

Two features of this principle make it resistant to formalization in a way that is *formal*, not merely sociological.

Obstruction 2.1 (non-finiteness). Each milestone quantifies over *all* height functions $f \rightarrow \infty$ and asserts existence of a density-one set, i.e. a statement of the shape $\forall f \exists (\text{good set}) \forall n \dots$ with an embedded limit $N \rightarrow \infty$. It is a genuine Π_3^0 -type arithmetical statement, not a Δ_1^0 decidable one. Consequently no `decide`, `native_decide`, or finite case-split can discharge it, and the exact finite identities of Proposition 1.3 (which *are* decidable) cannot be lifted by any purely computational tactic. A proof must construct the limit object.

Obstruction 2.2 (dependence). The increments are not independent as functions of x : $\mathbf{a}(\text{Syr}^i x)$ is determined by x , and `Syr` mixes residues in a j -dependent way. Independence holds only asymptotically, through equidistribution of the orbit modulo growing powers of 2. Formalizing the drift therefore cannot reuse an “i.i.d. law of large numbers”; one must first *prove* the equidistribution / coupling that manufactures the effective independence, together with a quantitative error that is uniform along the orbit. Mathlib has the abstract strong law (`ProbabilityTheory.strong_law_ae`) but nothing that turns a deterministic arithmetic sequence into an admissible input to it.

These two obstructions are already enough to explain why the milestones are not low-hanging: they are infinitary and they require building the very probabilistic model that makes the heuristics rigorous. The next three sections give the *additional*, milestone-specific obstructions.

3 Step 4: the density-one drift is a law of large numbers with no library

3.1 What must be produced

`step4_valuationLowerBound` demands, for every $f \rightarrow \infty$, a density-one set of starts n on which the valuation sum realizes the drift rate $8/5$ at *some* time $m \geq f(n)$:

$$8m \leq 5 \mathbf{S}_m(\text{oddn}), \quad m \geq f(n).$$

Since $\mathbb{E}[\mathbf{a}] = 2 > 8/5$, the “expected” inequality $8m \leq 5 \cdot 2m$ holds with room to spare; the content is that the *deterministic* sum $\mathbf{S}_m(\text{oddn})$ tracks $2m$ for almost all n , and does so at arbitrarily large prescribed times $m \geq f(n)$ simultaneously across a density-one set.

3.2 The three interlocking limits

Obstruction 3.1 (interchange of three limits). The statement couples three limiting processes that must be controlled *uniformly* against each other:

- (a) the number of steps $m \rightarrow \infty$ (over which the average $\mathbf{S}_m/m \rightarrow 2$ must concentrate);
- (b) the modulus $2^K \rightarrow \infty$ at which Proposition 1.3’s exact counts become the operative probability model, with K necessarily growing with m (an m -step prefix only equidistributes modulo 2^K once $K \gtrsim \mathbf{S}_m$);
- (c) the density scale $N \rightarrow \infty$ defining `DensityZero`, along which the exceptional set must vanish.

A naïve proof fixes each in turn and hopes the errors are summable; they are not, unless one has a *quantitative* large-deviation bound $\mathbb{P}[\mathbf{S}_m \leq \frac{8}{5}m] \leq e^{-cm}$ that is uniform in the coupling error between the arithmetic sequence and the geometric model. Producing that bound is Tao’s Proposition 1.9 in its quantitative, m -fold form — precisely the content that the finitary Proposition 1.3 does *not* give, because 1.3 is a first-moment/one-step statement.

3.3 Why the proved distribution does not suffice

Proposition 1.3 gives the exact one-step law and hence the mean $\mathbb{E}[\mathbf{a}] = 2$. A first instinct is: “mean $2 > 8/5$, so Markov/Chebyshev finishes it.” This fails formally for a specific, checkable reason.

Proposition 3.2 (first moments are not enough). *Let $g(n) :=$ the least $m \geq f(n)$ with $8m \leq 5\mathbf{S}_m(\text{oddn})$, or ∞ if none exists. The event “ $g(n) = \infty$ ” is the exceptional set of step 4. Its density cannot be bounded by any function of the one-step marginals of $\mathbf{a}$ alone: two orbit models with identical one-step marginal $\mathbb{P}[\mathbf{a} = j] = 2^{-(j+1)}$ but different joint correlation structure can have exceptional densities 0 and 1 respectively. Hence the proof must use the joint law of $(\mathbf{a}(\text{Syr}^i x))_i$, not merely Proposition 1.3.*

The informal justification is standard: concentration of $\mathbf{S}_m$ around its mean at an exponential rate is a statement about the *covariance decay* of the increments, which is invisible to the marginals. Formalizing step 4 thus forces one to establish decay of correlations for the deterministic sequence $\mathbf{a} \circ \text{Syr}^i$ — a mixing statement about the Syracuse map modulo 2^K that Mathlib does not contain in any usable form.

3.4 Missing infrastructure for step 4

Concretely, a Lean proof needs, and the library lacks: (i) a measure on the 2-adic integers $\mathbb{Z}_2$ (Mathlib has $\mathbb{Z}_2$ as a ring/topological space but not the Haar/valuation-pushforward measure with the needed geometric marginals); (ii) the pushforward of Lebesgue/counting measure under the Syracuse map and its equidistribution modulo 2^K with an effective rate; (iii) an exponential large-deviation inequality applicable to a *dependent, deterministic* sequence via that equidistribution; (iv) the transfer lemma from a $\mathbb{Z}_2$ -a.e. statement to a natural-density statement over $\mathbb{N}$. Each is a research-scale formalization in its own right.

4 Step 6: the reduction is provably not algebraic

Step 6 is the milestone whose difficulty is the most *surprising* and the one we can characterize most sharply, because here the obstruction is not “analysis is hard” but a *theorem*: the tempting purely-algebraic derivation is false.

4.1 The tempting derivation

The hypothesis `ContractionDominatesDensity` gives, on a density-one set, a time m with

$$3^m x \leq 2^{\mathbf{S}_m(x)}, \quad x = \text{oddn} \tag{1}$$

(taking $f \equiv 1$ inside the hypothesis; the actual hypothesis carries a height $f(n)$ but the point is unaffected). The desired conclusion is a strict descent $\text{Syr}^m(x) < x$ at some positive time. Using the exact backbone $2^{\mathbf{S}_m} \text{Syr}^m x = 3^m x + c_m$ (Lemma 1.2),

$$\text{Syr}^m(x) < x \iff 3^m x + c_m < 2^{\mathbf{S}_m} x \iff c_m < (2^{\mathbf{S}_m} - 3^m) x.$$

The “obvious” argument reads (1) as “ $2^{\mathbf{S}_m}$ dominates 3^m ” and concludes descent. *This inference is invalid*, and its failure is not a technicality: it is the reason step 6 needs the distribution.

4.2 The obstruction, precisely

Obstruction 4.1 (the additive constant is uncontrolled). The contraction hypothesis (1) constrains only the *multiplicative* pair $(3^m, 2^{\mathbf{S}_m})$ through the *sum* $\mathbf{S}_m$. But by Lemma 1.2 the additive correction

$$c_m(x) = \sum_{i=0}^{m-1} 3^{m-1-i} 2^{\mathbf{S}_i(x)}$$

depends on the *entire prefix profile* $(\mathbf{a}(x), \dots, \mathbf{a}(\text{Syr}^{m-1}x))$, not on $\mathbf{S}_m$ alone. Two starts with the same $\mathbf{S}_m$ (hence the same truth value of (1)) can have wildly different c_m : a profile that front-loads its valuation (large $\mathbf{a}$ early, then all 1’s) inflates c_m toward $2^{\mathbf{S}_m} x$, so that (1) holds while $\text{Syr}^m(x) \geq x$. Therefore (1) does not pointwise imply descent at that m .

We make the failure completely explicit.

Proposition 4.2 (explicit non-implication). *There is no constant C and no purely arithmetic function so that (1) implies $\text{Syr}^m(x) < x$ for the same m . Concretely, consider profiles with a large first valuation $\mathbf{a}(x) = A$ followed by minimal valuations. Then*

$$c_m(x) \geq 2^{\mathbf{a}(x)} c_{m-1}(\text{Syr}x) = 2^A c_{m-1}(\text{Syr}x),$$

which is $\Theta(2^{\mathbf{S}_m})$ as A grows with $\mathbf{S}_m$ fixed-fraction, whereas $(2^{\mathbf{S}_m} - 3^m)x$ stays $\Theta(2^{\mathbf{S}_m}x)$; choosing the profile so the first swamps the second gives (1) with $\text{Syr}^m(x) \geq x$. Hence descent must be sought at a different time than the one (1) advertises, and selecting that time is a distributional statement about which prefixes are typical.

Remark 4.3. This is exactly the degeneracy the milestone’s docstring warns against: the trivial $m = 0$ reading ($\text{Syr}^0 x = x$) and, more subtly, the “wrong- m ” reading above. The Lean statement forbids the former ($0 < m$ required); the latter is why `step6_typicalDescent` cannot be closed by `omega/nlinarith` from its hypothesis plus the backbone, however much arithmetic automation is thrown at it.

4.3 What a correct proof must instead do

Because the pointwise implication fails, step 6 must argue: *for almost all x , among the times m where (1) holds there is one at which the additive constant is small*, i.e. $c_m(x) < (2^{\mathbf{S}_m} - 3^m)x$. Controlling c_m requires knowing that the prefix profile of a typical x is not front-loaded — again a statement about the joint distribution of the valuations (the increments are typically “spread out”, so $\mathbf{S}_i \approx 2i$ for all $i \leq m$, which makes $c_m = \sum 3^{m-1-i}2^{\mathbf{S}_i}$ geometrically dominated by its last term). Thus step 6, despite looking like an inequality manipulation, reduces to the same concentration-of-the-profile content as step 4, applied *uniformly along the prefix* rather than only at the endpoint. This is why it is a separate, genuinely analytic hole and not a corollary of step 5.

5 Step 7: self-referential density and the renewal obstruction

Step 7 upgrades a *single* typical strict descent below the start to density-one *first passage below an arbitrary height* $f \rightarrow \infty$:

$$\text{TypicalDescentDensity} \implies \text{SyracuseDensityControl}.$$

5.1 Why one descent is not enough, and why iteration is subtle

From `TypicalDescentDensity` we know a density-one set G of starts x has *some* $m > 0$ with $\text{Syr}^m(x) < x$. To reach below a fixed height $f(n)$ one must iterate: descend to $x_1 < x_0$, then to $x_2 < x_1$, and so on until below $f(n)$. The obstruction is that iteration requires the descent property to hold *again at the descendant* $x_1 = \text{Syr}^m(x_0)$, and x_1 need not lie in the original good set G .

Obstruction 5.1 (non-inheritance of the good set). Density-one-ness is a property of a set of *starts*, measured by natural density over $\mathbb{N}$. The Syracuse map does *not* preserve natural density in either direction in an elementary way: the image $\text{Syr}(G)$ of a density-one set G carries no a priori density lower bound, and the pre-image structure is governed by the same equidistribution one is trying to establish. Hence “ $x_0 \in G$ ” gives no information about “ $x_1 \in G$ ”. Iterating the descent therefore cannot be a straight induction; the good set must be redesigned to be *stable under the dynamics*.

5.2 The renewal / stopping-time content

The correct object is a *renewal* argument (Tao’s Propositions 1.11 / 7.8): one defines first-passage / stopping times $\tau_k(x) = \min\{m : \text{Syr}^m(x) < x/2^k\}$ and shows that the exceptional set at each renewal epoch has controlled density, *summably* over k , so that a density-one set survives all epochs simultaneously. Formalizing this needs:

Obstruction 5.2 (quantitative renewal with summable exceptional sets). One must prove a *stability estimate*: the density of starts that fail to descend by a further factor within a bounded number of steps decays fast enough in the descent depth k that $\sum_k$ (exceptional density at depth k) is finite and the overall exceptional set has density zero. This is a Borel–Cantelli-type argument over the dynamically-generated σ -algebra of first-passage events, requiring: (i) a renewal-process formalism (first-passage times of a Markov-like deterministic system) that Mathlib

does not have; (ii) monotonicity of the tail estimate under passage to descendants (the proved `syrVal_level_count_period` gives the base tail but not its stability along the orbit); and (iii) a diagonalization matching the renewal depth k to the growing height $f(n)$ uniformly.

5.3 The logical shape of the difficulty

Step 7 is where the statement becomes genuinely *self-referential*: the hypothesis and conclusion are both density-one assertions about the *same* dynamical system, and the proof must feed the conclusion back into itself along the orbit. This is structurally a fixed-point/coinductive argument dressed as a renewal theorem, and it is the reason step 7 cannot be a finite reduction: there is no bound on how many descent epochs are needed to get below $f(n)$, and each epoch consumes a fresh instance of the (analytic) descent guarantee. Even granting steps 4 and 6 as black boxes, assembling them into `SyracuseDensityControl` is a non-trivial infinitary bookkeeping problem.

6 The library gap, tabulated

The three obstruction families translate into concrete missing Mathlib infrastructure. The table records, for each needed component, whether Mathlib currently provides it in a form usable for these milestones.

Needed component	In Mathlib?	Gap
Measure on $\mathbb{Z}_2$ with valuation marginals $\mathbb{P}[\mathbf{a} = j] = 2^{-(j+1)}$	partial	$\mathbb{Z}_2$ exists as a ring/space; the push-forward valuation measure and its marginals are not developed.
Equidistribution of Syr-orbits mod 2^K with an effective rate	no	no formalization of Syracuse-orbit equidistribution; general equidistribution API is thin and not applicable to this deterministic map.
Large-deviation / concentration for <i>dependent, deterministic</i> sequences	partial	abstract i.i.d. strong law (<code>strong_law_ae</code>) exists; no bridge from an arithmetic sequence + mixing to its hypotheses.
Correlation-decay (mixing) of $\mathbf{a} \circ \text{Syr}^i$	no	needed for Prop. 3.2/step 4; absent.
Renewal-process / first-passage formalism	no	needed for step 7; no first-passage or renewal library.
Natural-density $\leftrightarrow \mathbb{Z}_2$ -a.e. transfer	no	needed to convert measure statements to <code>DensityZero</code> ; absent.
Density interchange with a growing modulus $K = K(m, N)$	no	the uniform triple-limit control of Obstruction 3.1.

The Idris2 side of the development builds a *minimal* rigorous model of the first four rows (the `FinDist` finitely-supported-measure carrier, with machine-checked Markov, convolution/Fourier, and renewal power laws, and a genuine 2-adic-valuation instance). That model is exactly a scaffold: it makes the interface nodes non-vacuous and proves the algebraic backbone of the four analytic domains, but it deliberately stops short of the quantitative estimates that discharge the milestones — which is the honest statement of the boundary.

7 Summary: three different reasons, one honest boundary

We can now state the characterization compactly.

Theorem 7.1 (characterization of the three obstructions). *Each remaining milestone is irreducible to the proved arithmetic for a distinct, identifiable reason:*

- (1) **step 4** is a quantitative law of large numbers for a deterministic, dependent arithmetic sequence, requiring the joint valuation distribution (Obstruction 2.2), a uniform triple-limit interchange (Obstruction 3.1), and infrastructure (a $\mathbb{Z}_2$ measure, equidistribution, concentration) that the library lacks. It is not finite (Obstruction 2.1) and not implied by the proved one-step marginals (Prop. 3.2).
- (2) **step 6** fails to be a pointwise algebraic reduction: the exact backbone's additive constant c_m depends on the whole prefix profile and is not controlled by the contraction hypothesis (Obstruction 4.1), so the tempting derivation is provably invalid (Prop. 4.2); a correct proof needs the same concentration content as step 4, applied along the whole prefix.
- (3) **step 7** is self-referential: it must inherit its own density hypothesis along the orbit's descendants (Obstruction 5.1) via a quantitative renewal argument with summable exceptional sets (Obstruction 5.2), an infinitary assembly with no library support.

In every case the obstruction survives the removal of all surrounding arithmetic: the elementary dynamics, the exact affine backbone, the exact one-step valuation distribution, and the entire reduction chain are proved without axioms, and none of them shortens the gap.

The value of the formalization is precisely that it *localizes* the difficulty. Tao's theorem is not opaque here: it has been peeled down to three statements whose hardness is not a matter of opinion but of structure — an infinitary probabilistic transfer (step 4), a provable failure of algebraic reducibility that forces a distributional argument (step 6), and a self-referential renewal (step 7). These are the genuine analytic heart of the paper, and this note has given, for each, a rigorous account of why it cannot be discharged by the finite, elementary, or automated means that suffice for everything around it.